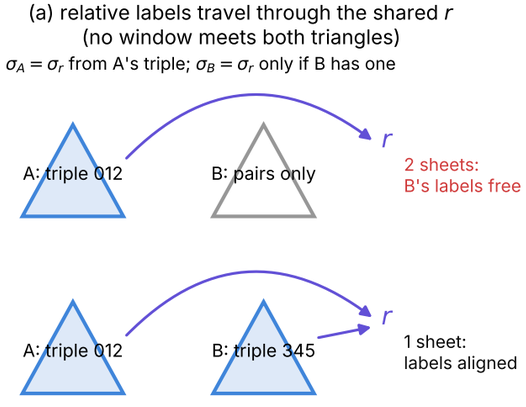
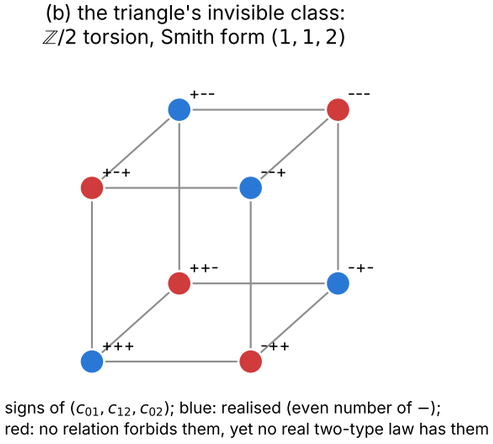
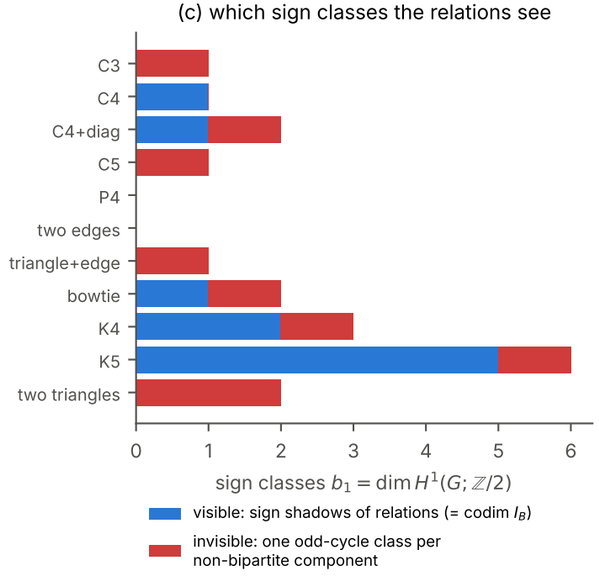
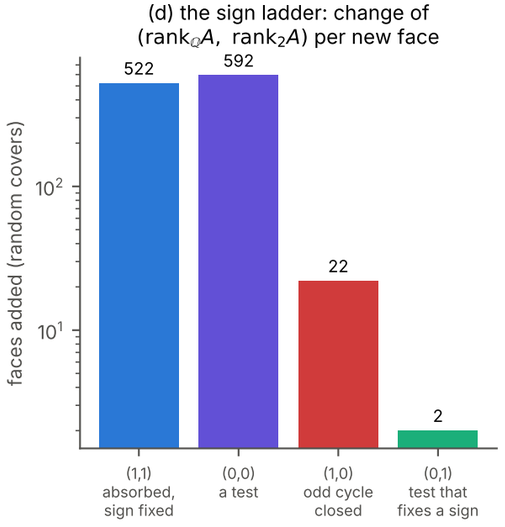

(a) relative labels travel through the shared r
(no window meets both triangles)
$\sigma_A = \sigma_r$ from A's triple; $\sigma_B = \sigma_r$ only if B has one
A: triple 012
B: pairs only
r
2 sheets:
B's labels free
A: triple 012
B: triple 345
r
1 sheet:
labels aligned


(b) the triangle's invisible class:
$\mathbb{Z}/2$ torsion, Smith form (1, 1, 2)
+--

+-+
-++
++-
-+-
+++
-++
signs of (c_{01}, c_{12}, c_{02}); blue: realised (even number of −);
red: no relation forbids them, yet no real two-type law has them


(c) which sign classes the relations see
C3
C4
C4+diag
C5
P4
two edges
triangle+edge
bowtie
K4
K5
two triangles
0
1
2
3
4
5
6
sign classes $b_1 = \dim H^1(G; \mathbb{Z}/2)$
visible: sign shadows of relations (= codim I_B)
invisible: one odd-cycle class per
non-bipartite component


(d) the sign ladder: change of
($\mathrm{rank}_{\mathbb{Q}} A$, $\mathrm{rank}_2 A$) per new face
faces added (random covers)
10^2
10^1
522
592
22
2
(1,1)
absorbed,
sign fixed
(0,0)
a test
(1,0)
odd cycle
closed
(0,1)
test that
fixes a sign